آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-I Business Mathematics Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of the Higher Secondary School Certificate (HSSC) Part I Business Mathematics. It contains comments on candidates' responses to the 2025 HSSC-I Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding, and application skills they have developed during study. Candidates are advised to read and comprehend each question carefully before writing their response to fulfill the question's demand.

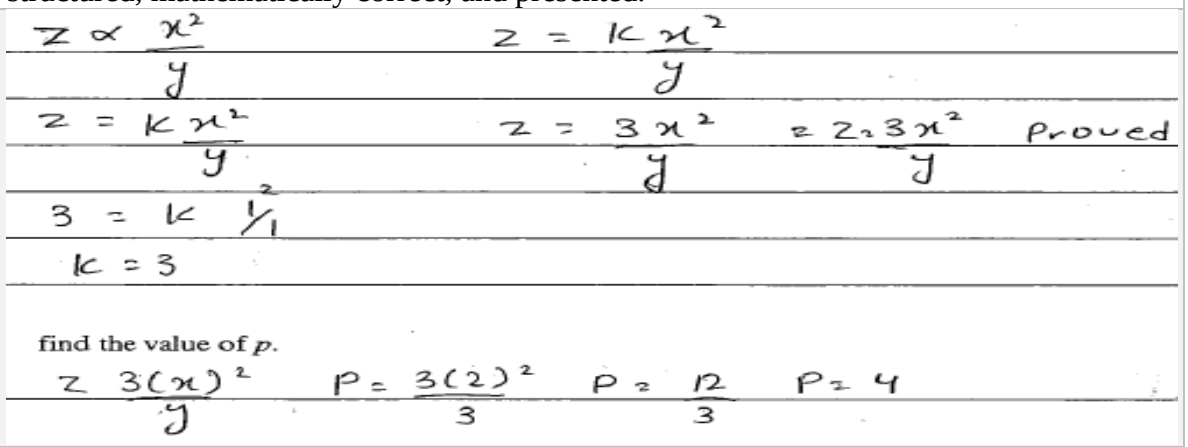
Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs, which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why', or 'what' may also be used. It is imperative to refer to the command word guide available on AKU-EB website for understanding the expectation of the command word.

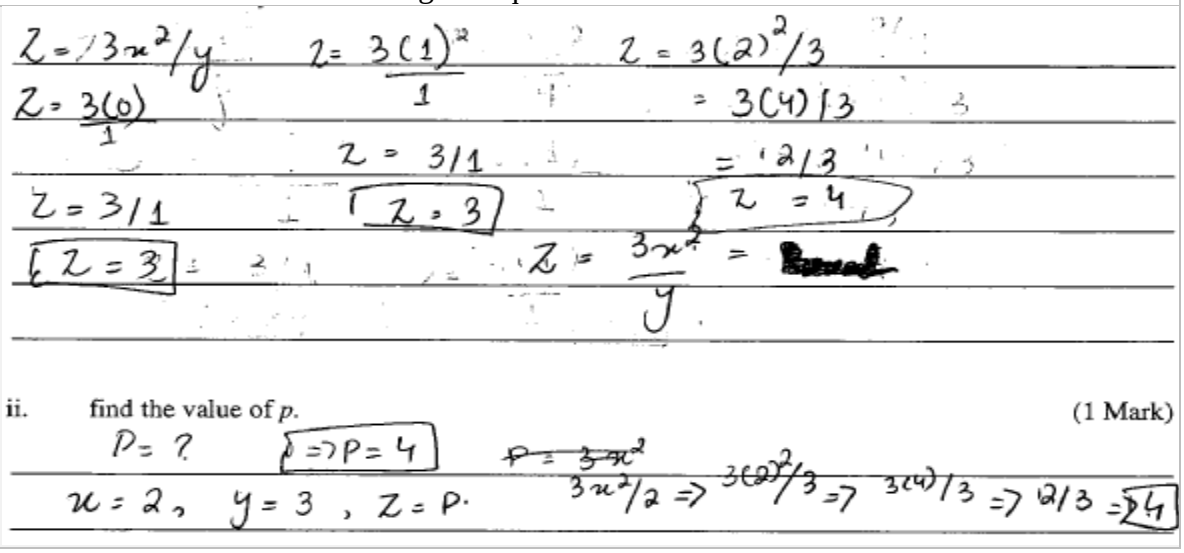
General Observations

Overall, candidates performed well on questions involving **compound interest**, **simultaneous linear equations**, **graphs of quadratic functions**, **matrices**, and **arithmetic series**. Their responses reflected a solid understanding of the underlying concepts and the correct application of relevant methods and formulas. However, there were noticeable difficulties in areas such as **compound variation**, **annuities**, **linear inequalities and shading the correct region**, and **geometric series**. These topics often revealed gaps in conceptual understanding, improper use of formulae, and procedural errors, indicating the need for further reinforcement and targeted practice in these specific areas.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

<table>
<tr><th colspan="2">Question No. 1</th></tr>
<tr><td>Question Text</td><td>The relationship between three quantities x, y and z is that z varies directly with x^2 and inversely with y.
Their relationship is shown in the given table.
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td></tr>
<tr><td>y</td><td>1</td><td>1</td><td>3</td></tr>
<tr><td>z</td><td>0</td><td>3</td><td>p</td></tr>
</table>
Using the given information,
i. prove that $z = \frac{3x^2}{y}$.
ii. find the value of p.</td></tr>
<tr><td>SLO No.</td><td>1.2.2</td></tr>
<tr><td>SLO Text</td><td>Solve word problems related to SLO 1.2.1.
(1.2.1. Describe direct, inverse and compound (joint and combined) variations.)</td></tr>
<tr><td>Max Marks</td><td>4</td></tr>
<tr><td>Cognitive Level</td><td>*A</td></tr>
<tr><td>Checking Hints</td><td>i. 1 mark for writing $z \propto x^2$
1 mark for writing $z \propto \frac{1}{y}$
Or
2 marks for writing $z \propto \frac{x^2}{y}$
1 mark for finding the value of k
ii. 1 mark for calculating the value of p</td></tr>
<tr><td>Overall Performance</td><td>Overall, performance on this question was diverse. While some candidates represented a sound understanding of proportional relationships and followed the correct method to solve the problem, others had difficulty identifying and applying the correct approach. Errors were mostly related to misinterpreting the question, substituting values incorrectly, or not forming the proportional equation as required.</td></tr>
<tr><td>Description of Better Responses</td><td>In better responses, the candidate correctly interpreted the proportional relationship by forming the appropriate equation. They represented a clear understanding of both direct and inverse proportionalities. Using the given values of Z, x, and y, the candidates substituted them into the equation and accurately solved for the constant k. After finding the value of k, they substituted it back into the original expression and verified the relationship by showing that the equation held under the given conditions. Their working was logically structured, mathematically correct, and presented.</td></tr>
<tr><td>Image of Better Response</td><td> 
</td></tr>
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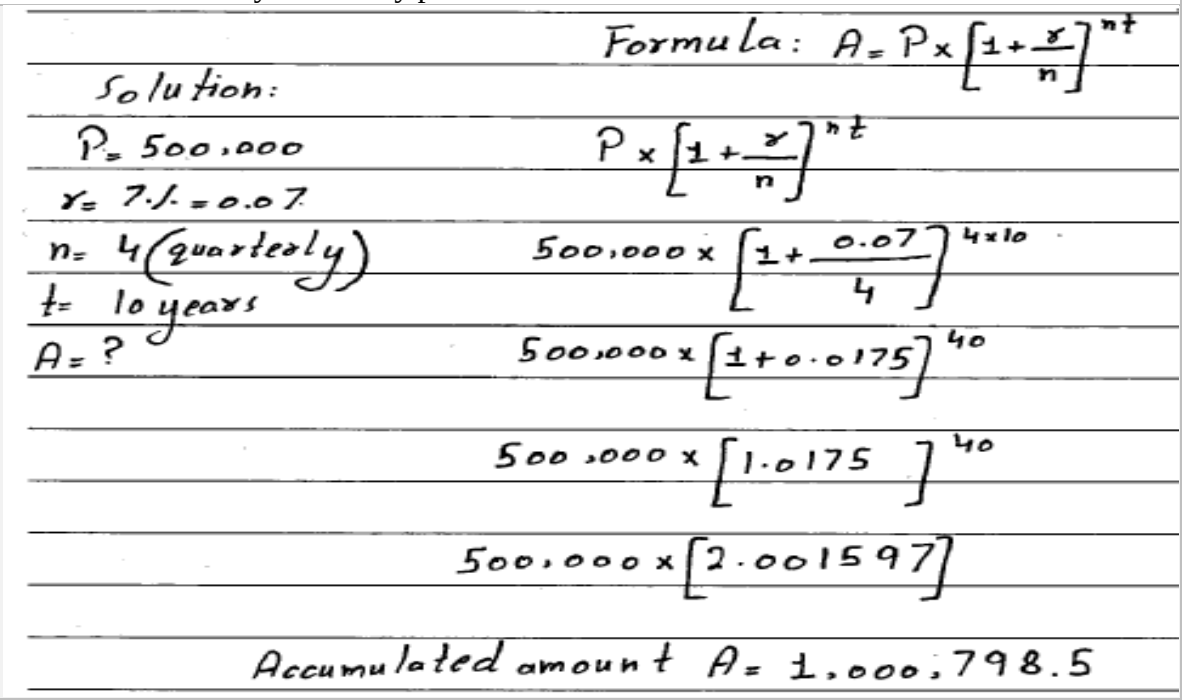
Description of Weaker Responses	In *weaker responses*, candidates struggled to interpret the relationship described in the question. Instead of writing the correct proportional equation, they directly substituted the values of *x* and *y* from the given table and tried to show that the values matched *Z*. Common issues included the misuse of variables, incorrect substitution, and a lack of understanding of the correct method for forming the equation.
Image of Weaker Response	

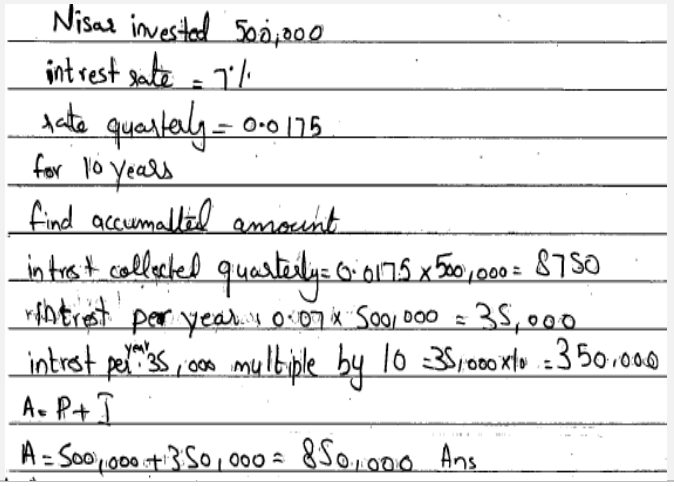
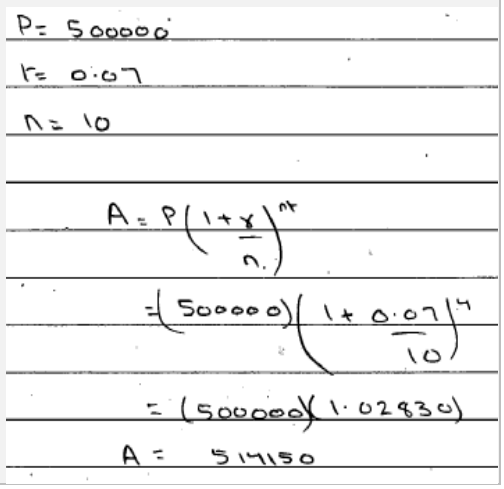
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
By implementing these strategies, educators can enhance students' understanding and application of complex concepts, ultimately improving overall performance on assessments.
Real-Life Context Problems: Use real-life examples (e.g., speed and time, work and number of workers, price and quantity) to help candidates relate to the concept.
Strategy: Ask candidates to identify whether the relationship in each example is direct or inverse, then write the proportional equation.
Matching Card Activity: Prepare sets of cards with proportional equations, graphs, tables, and verbal descriptions. Candidates match the correct sets together.
Strategy: Encourage group work to promote discussion and reinforce understanding of the connection between different representations.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2a Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.	
Question Text	Nisar invested Rs 500,000 in a savings account with a fixed annual interest rate of 7% compounded quarterly. Find the accumulated amount after 10 years.
SLO No.	2.2.5
SLO Text	Solve word problems related to the concept of compound interest.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for finding the value of *r* 1 mark for identifying the number of compounding periods per year (quarterly) 1 mark for using the correct formula 1 mark for calculating the accumulated amount
Overall Performance	This question was selected more by the candidate than part b. Overall, performance on this question good. While many candidates represented a clear understanding of compound interest and applied the formula correctly, few demonstrated gaps in their understanding of how compounding works, particularly about rate and period adjustments. More practice on distinguishing between simple and compound interest, as well as careful substitution and use of brackets, is recommended to improve accuracy.
Description of Better Responses	In *better responses*, candidates correctly identified the compound interest formula and understood the role of the compounding period. They accurately substituted the values of rate, time, and period into the formula and correctly converted the percentage rate into decimal form. They also handled the compounding frequency appropriately, such as recognising that quarterly means a frequency of 4. Their final answers were accurate, and the work was clearly and neatly presented.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, some candidates struggled to apply the compound interest formula. Instead of using the compound interest structure, they misinterpreted the frequency of compounding. In some cases, the candidates used the simple interest formula, inserted incorrect values, or skipped necessary brackets during exponentiation. A common mistake was failing to divide the interest rate by the number of compounding periods, resulting in incorrect results.

Images of Weaker Responses	Image (i)	Image (ii)
		

Suggestions for improvement (Highlight all that apply)

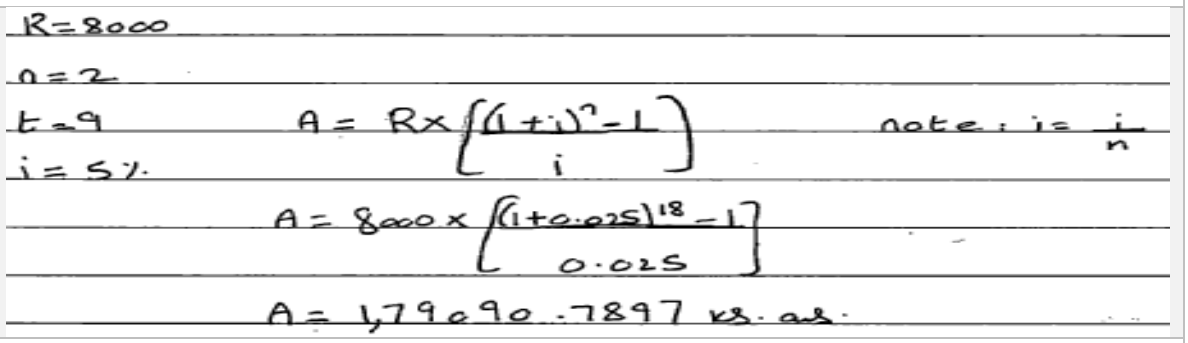
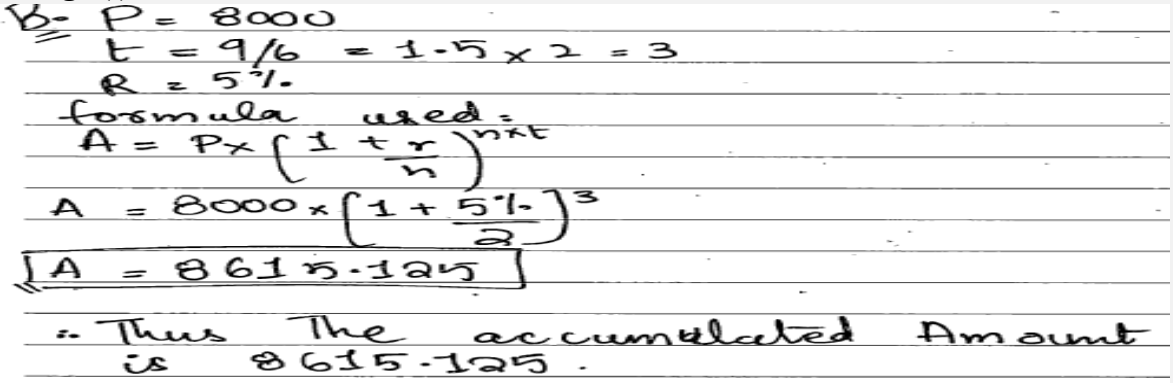
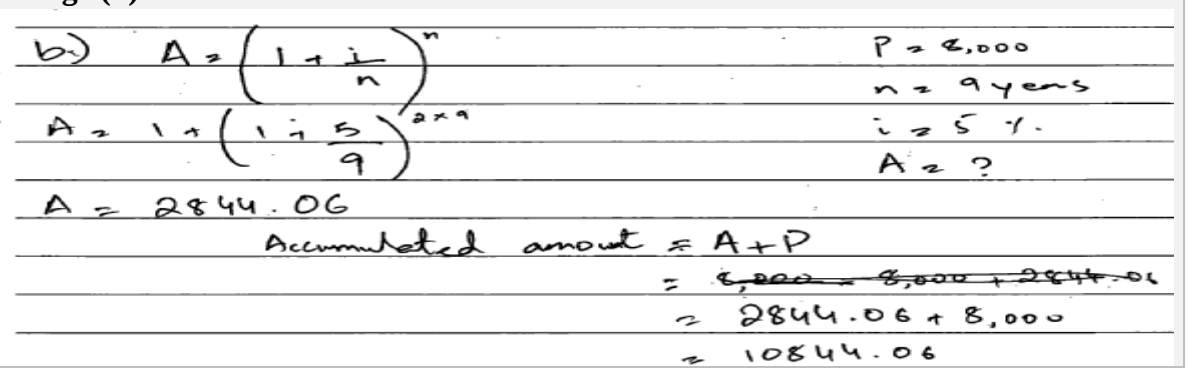
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:
To enhance candidate's performance in this concept teachers may consider the following strategies:
Interest Growth Timeline: Have candidates draw a timeline showing the growth of an investment or loan over several periods (e.g., yearly or quarterly).
Strategy: Helps visualise how interest is added to the principal each time and how the amount grows.
Simple vs Compound Interest: Organise a small group debate or discussion where candidates compare the pros and cons of both interest types in real-life situations.
Strategy: Encourages critical thinking and real-world application.
Interactive Online Tools: Use free tools such as compound interest calculators or math learning platforms (like Desmos, GeoGebra, or Mathigon).
Strategy: Encourages independent learning and helps visual learners.

Question No. 2b

Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.

Question Text	Zeeshan deposited Rs 8,000 at the end of each period of six months for 9 years at 5% interest compounded semi-annually. Find his accumulated amount.
SLO No.	2.3.5
SLO Text	Solve word problems related to the concept of ordinary annuity i.e. investment, deposit and loans.
Max Marks	4

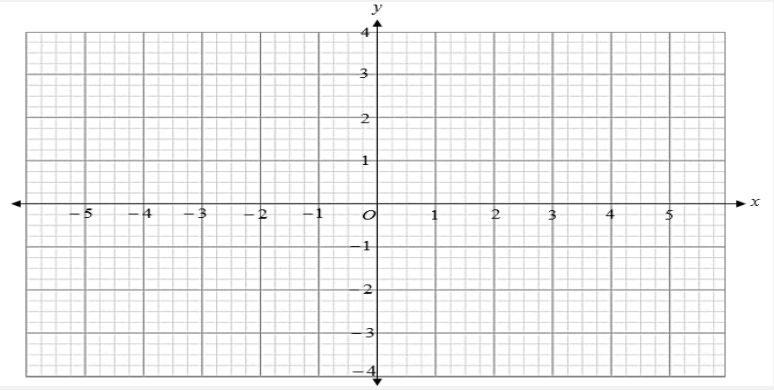
Cognitive Level	A
Checking Hints	1 mark for correct value of *n* 1 mark for correct value of *r* 1 mark for using the correct formula 1 mark for calculating the accumulated amount
Overall Performance	Very few candidates attempted this part of the question. Among those who did, performance varied. While some candidates correctly applied the annuity formula and reached the expected answers, many had difficulty identifying key components such as the interest rate, number of periods, and correct substitution into the formula.
Description of Better Responses	In *better responses*, candidates showed a clear understanding of the annuity concept. They correctly identified the values for the regular payment amount, interest rate, time, and compounding frequency. They substituted these values accurately into the annuity formula and carried out the calculations properly. Their work was clear, and their use of the method led to the correct final answer.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates struggled to understand how annuities work. Some confused the annuity formula with the compound interest formula, while others were unsure how to interpret the interest rate or the number of compounding periods. A few candidates misread the values provided in the question or had difficulty using the variables listed on the formula sheet. This resulted in incorrect substitutions, calculation errors, and incomplete solutions.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:
To enhance overall performance, teachers may consider the following strategies:
Timeline-Based Calculation Practice: Use visual timelines for multi-period payments to show how each payment accumulates interest.
Activity: Have students label payment dates, calculate interest per payment, and track totals manually before using the formula.
Real-Life Scenario Analysis: Use practical examples such as retirement savings, loan repayments, or monthly investments.
Activity: Ask students to calculate how much they need to save monthly to reach a specific goal in a given number of years.

Question No. 3

Question Text	The given table contains the points satisfying the function. x: −1, 0, 1 y: 1, 0, 1 i. Using the given table, sketch the graph of the function.  ii. On the graph, identify the vertex.
SLO No.	4.2.3 and 4.2.6
SLO Text	Find the vertex, intercepts, axis of symmetry and direction (concavity) of a quadratic function. Sketch the graph of a quadratic functions, i.e. parabola.
Max Marks	3
Cognitive Level	A

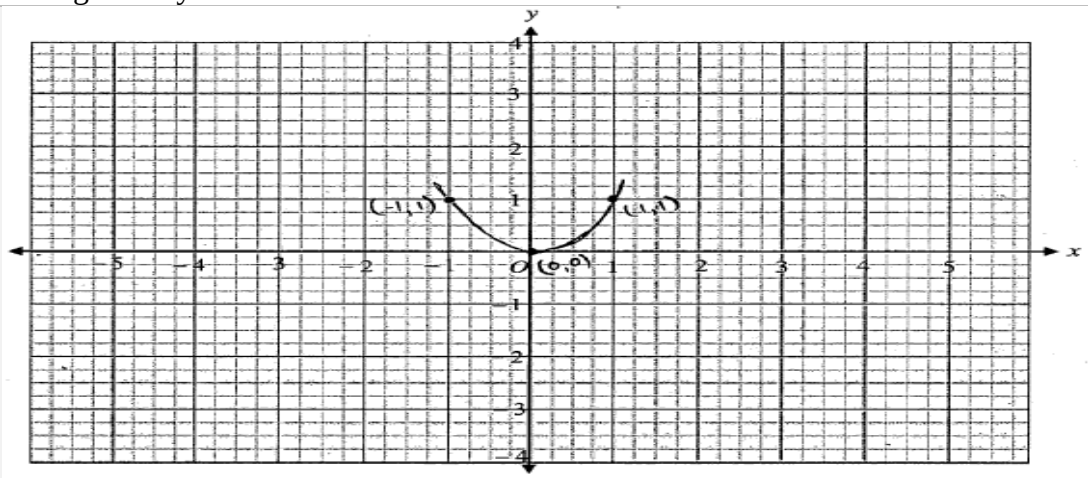
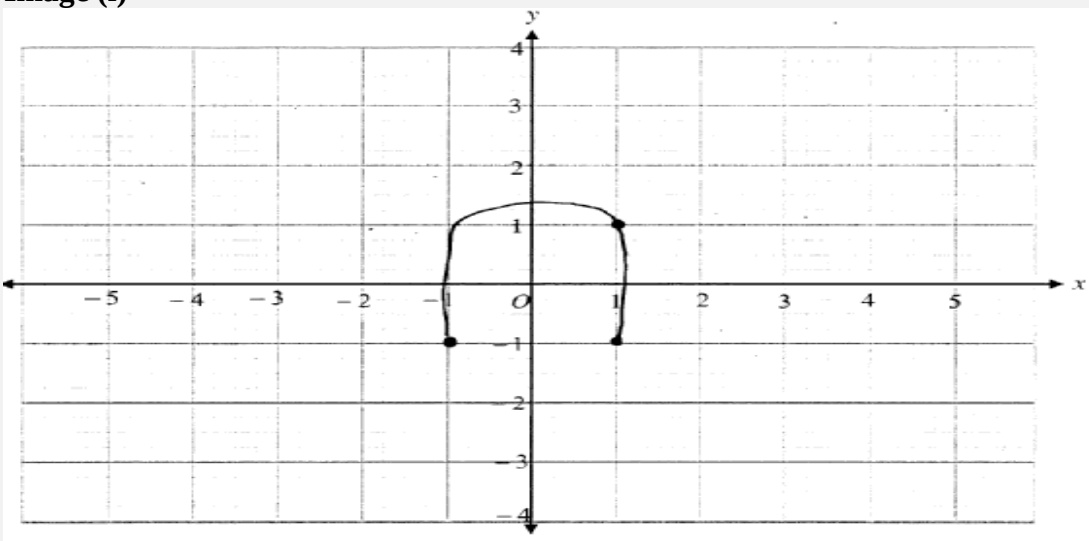
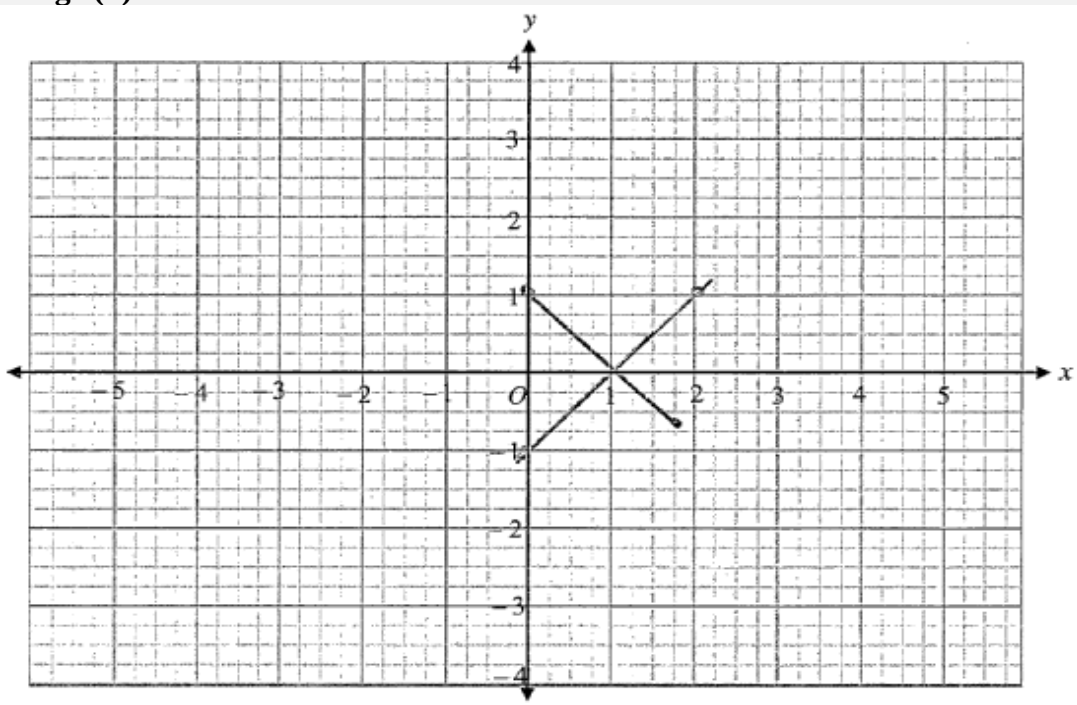
Checking Hints	i. 1 mark for plotting the points 1 mark for the drawing the graph correctly ii. 1 mark for identifying the vertex of the function (0, 0)
Overall Performance	Overall, performance on this question was good. many candidates demonstrated a good understanding of the graphical representation of a quadratic function and correctly identified the vertex. However, few struggled with distinguishing between linear and quadratic graphs. Errors in plotting points, misusing formulas, or misunderstanding the concept of a turning point suggest a need for more focused practice on recognising and interpreting the shape and features of a quadratic graph.
Description of Better Responses	In *better responses*, candidates were able to draw the correct graph of the quadratic function and correctly identify the vertex from the graph. They demonstrated an understanding that the vertex is the turning point of the parabola and used this to find key values. Their graphs were generally accurate and well-labelled.
Image of Better Response	 ii. On the graph, identify the vertex. (1 Mark) The vertex on the graph is the point (0,0)
Description of Weaker Responses	In *weaker responses*, some candidates were unable to draw the quadratic function correctly. A few drew incorrect parabolas, while others mistakenly drew straight lines, indicating confusion between quadratic and linear functions. Many did not understand that the vertex is the turning point of the parabola, which can be identified visually from the graph. Some applied the wrong values in the vertex formula or misinterpreted the graph, which led to incorrect answers.
Images of Weaker Responses	**Image (i)**  ii. On the graph, identify the vertex. (1 Mark) $\frac{-b}{2(a)} = \frac{-(0)}{2(-1)} = 0$ Point of origin

Image (ii)



ii. On the graph, identify the vertex. (1 Mark)

The vertex is O.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

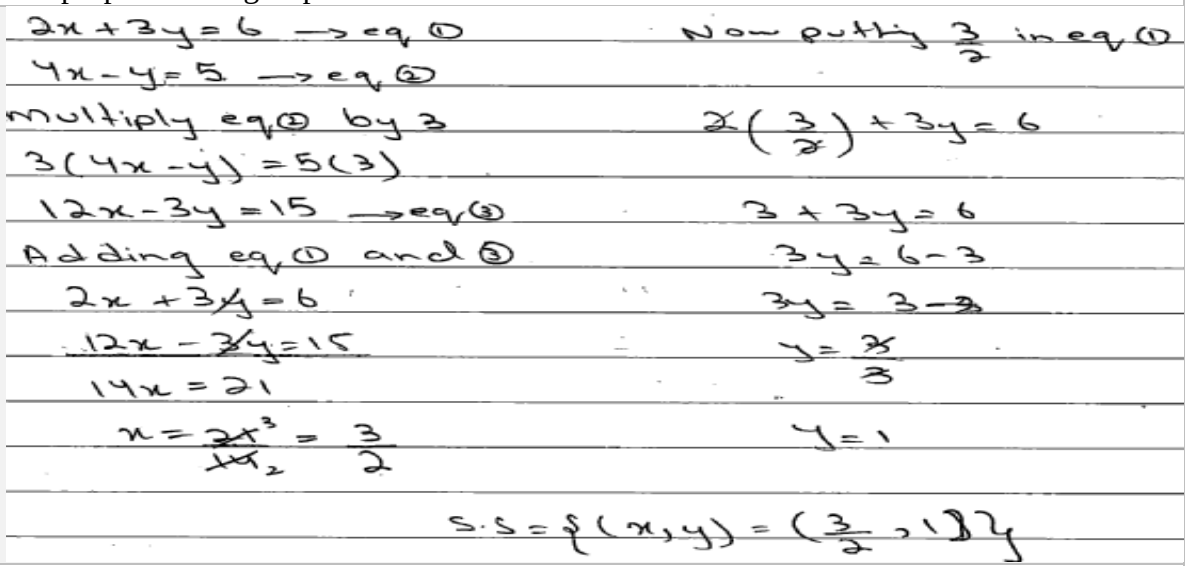
To enhance the performance of this concept, teachers may consider the following strategies:

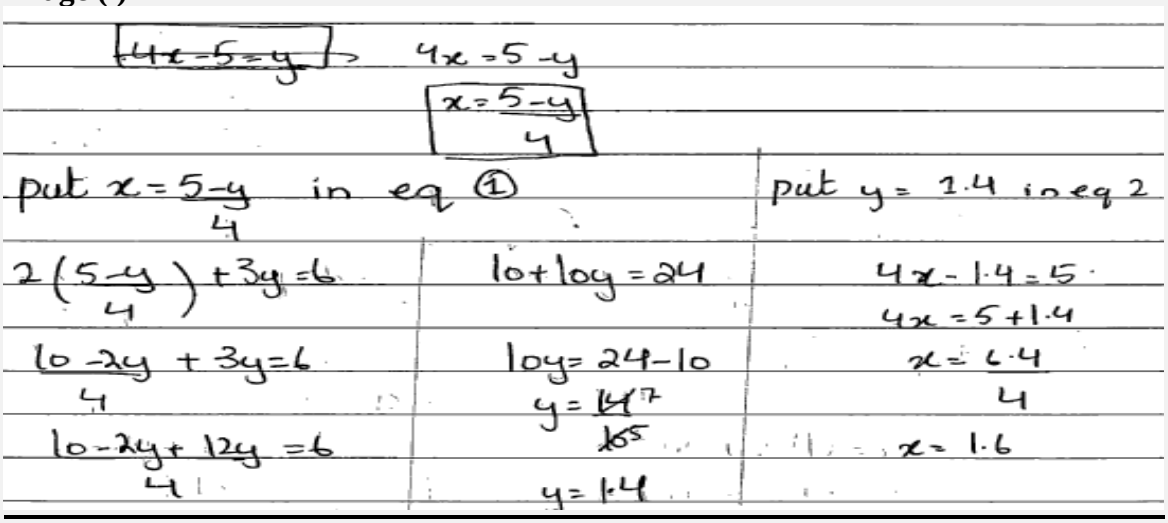
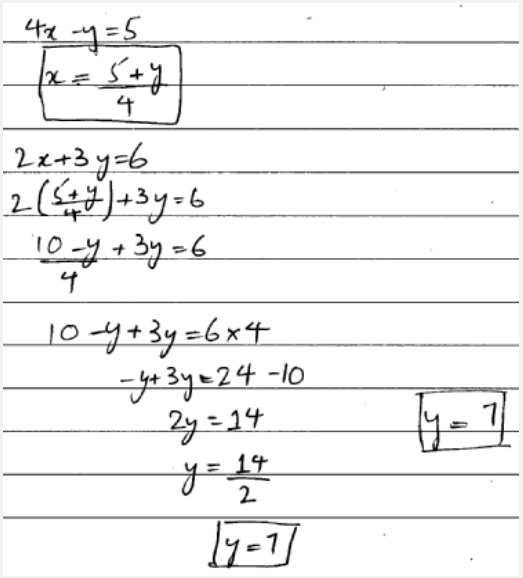
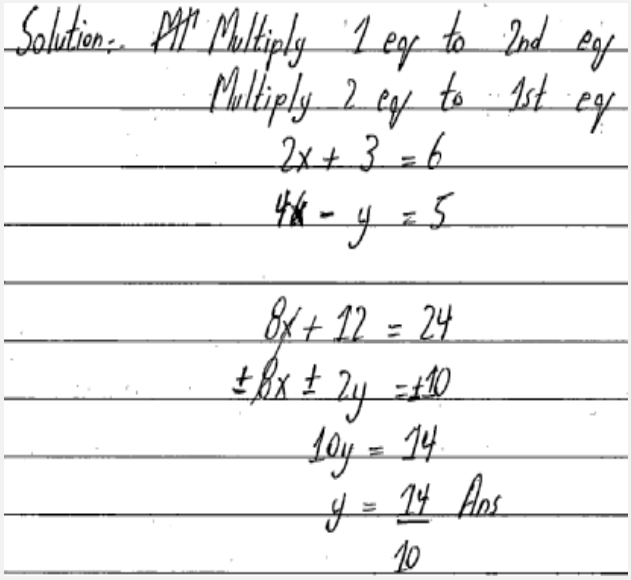
Use of Graphing Technology: Utilise tools like Desmos, GeoGebra, or graphing calculators to visually illustrate how modifying the coefficients in $ax^2 + bx + c$ affects the shape and position of the parabola.

Activity: Have students experiment by changing values of a, b, and c, then describe how the graph shifts or changes direction.

Graphing from a Table of Values: Teach students to generate a table of x and y values for given quadratic functions before plotting.

Activity: Assign practice problems where students complete the table, plot points, and connect them to form the parabola.

Question No. 4a Candidates were given the choice to attempt any ONE out of the two questions: 4a and 4b.	
Question Text	Find the solution of the following simultaneous linear equations. $2x+3y=6 \qquad 4x-y=5$
SLO No.	5.1.1
SLO Text	Solve simultaneous equations: a. both are linear b. both are quadratic, i.e., $ax^2 + by^2 = m \; ; \; cx^2 + dy^2 = n$ c. one is linear and other is quadratic, i.e., $ax + by = m \; ; \; cx^2 + dy^2 = n$.
Max Marks	3
Cognitive Level	A
Checking Hints	**For elimination method** 1 mark for finding the value of $x=\frac{3}{2}$ or $y=1$ 1 mark for substituting the value of *x or y* in one of the equations 1 mark for finding the value of *y* or *x* **For substitution method** 1 mark for making equation (iii) 1 mark for finding the value of *x or y* in one of the equations 1 mark for finding the value of *y* or *x*
Overall Performance	The majority of candidates attempted this part of the question. Performance was also as per the expectations. While many candidates demonstrated a clear understanding of solving simultaneous equations using either the substitution or elimination method, few candidates struggled with the basic calculations and algebraic manipulation required. Common issues included incorrect substitution**,** multiplication errors, and difficulty in isolating variables correctly.
Description of Better Responses	In *better responses*, some candidates used the substitution method, while others applied the elimination method to solve the pair of simultaneous equations. Both methods were used correctly. Candidates were able to find the value of one variable from one equation and accurately substitute it into the other equation to find the second variable. The final answer was presented clearly in solution set form, showing a good understanding of the method and proper working steps.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, several candidates made calculation errors, which reflected weaknesses in their basic mathematical skills. Many were unable to correctly substitute values from one equation into the other, leading to incorrect results. Some attempted the

	elimination method but performed the multiplication step incorrectly, which affected the rest of their solution. As a result, they failed to find the correct values of *x* and *y*.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

To enhance the performance of this concept, teachers may consider the following strategies:

Method Selection Quiz

Students are given several pairs of equations and are asked to choose whether substitution or elimination is the better method for each. No solving is required, just method selection. This builds decision-making skills and helps students become more strategic in their approach to solving equations.

Colour-Coding Variables

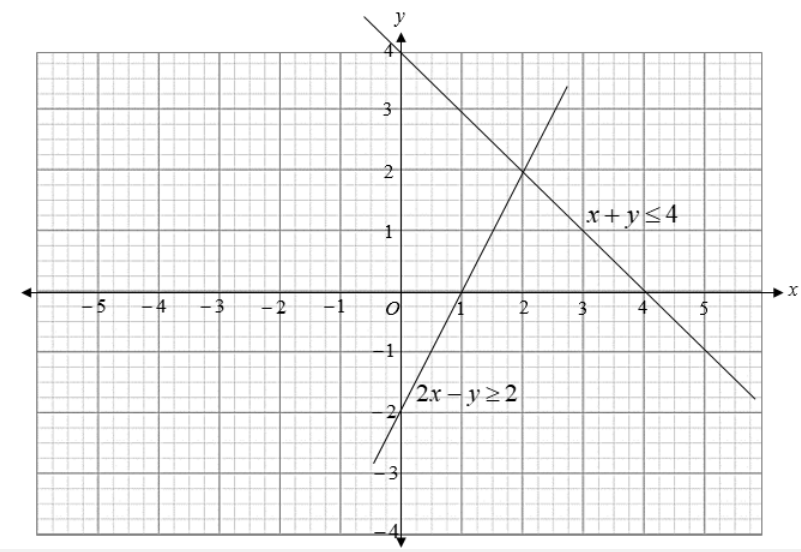
Students use different coloured pens or highlighters to mark x and y terms in both equations before solving. This simple visual aid helps reduce confusion between variables and enables students to focus on the correct terms during substitution or elimination, making it especially useful for visual learners.

Mistake Hunt

Provide students with worked examples that contain intentional errors (e.g., sign mistakes, incorrect substitution, wrong multiplication). Students identify the errors and correct them. This develops critical thinking and helps students become better at checking their work during assessments.

Question No. 4b

Candidates were given the choice to attempt any ONE out of the two questions: 4a and 4b.

Question Text	Consider the given linear programming problem. $x + y \leq 4 \qquad 2x - y \geq 2 \qquad x \geq 0,\ y \geq 0$ On the given graph, i. shade the feasible region.  ii. identify all the corner points of the feasible region.
SLO No.	5.2.4
SLO Text	Solve problems based on linear inequalities in two variables.
Max Marks	3
Cognitive Level	U
Checking Hints	i. 1 marks for shading the correct region ii. 1 mark for identify the two correct corner points of the feasible region (2required)
Overall Performance	This question was not attempted by most. Even those who attempted it the performance on this question was not up to the mark. While few candidates demonstrated a good understanding of shading regions and identifying feasible areas, many showed confusion in interpreting inequality conditions and finding points of intersection. Errors in shading and selecting the correct region suggest that more practice is needed in graphically representing linear inequalities and identifying the feasible region and its corner points accurately.

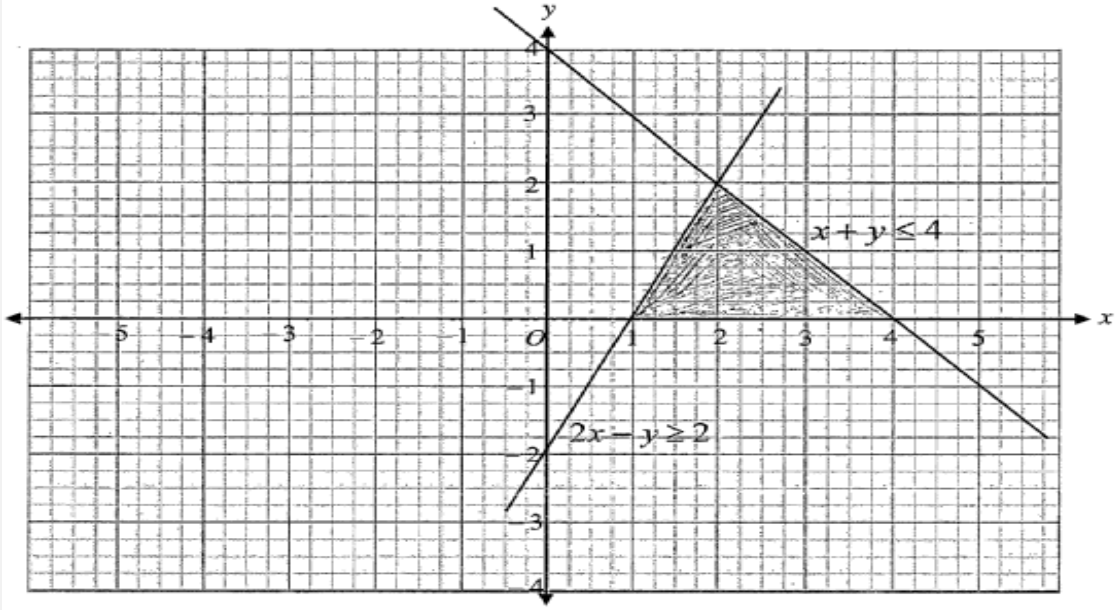
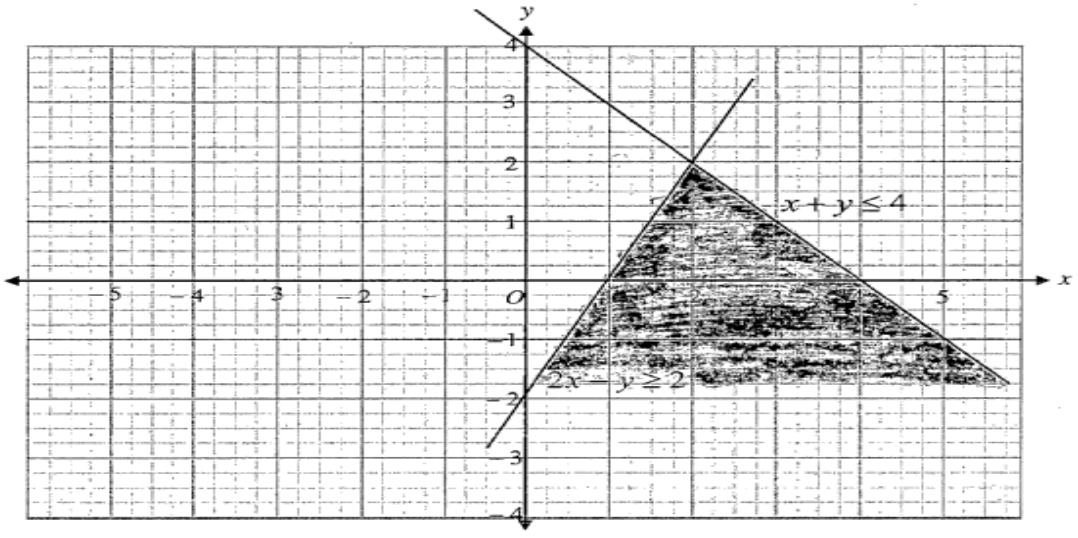
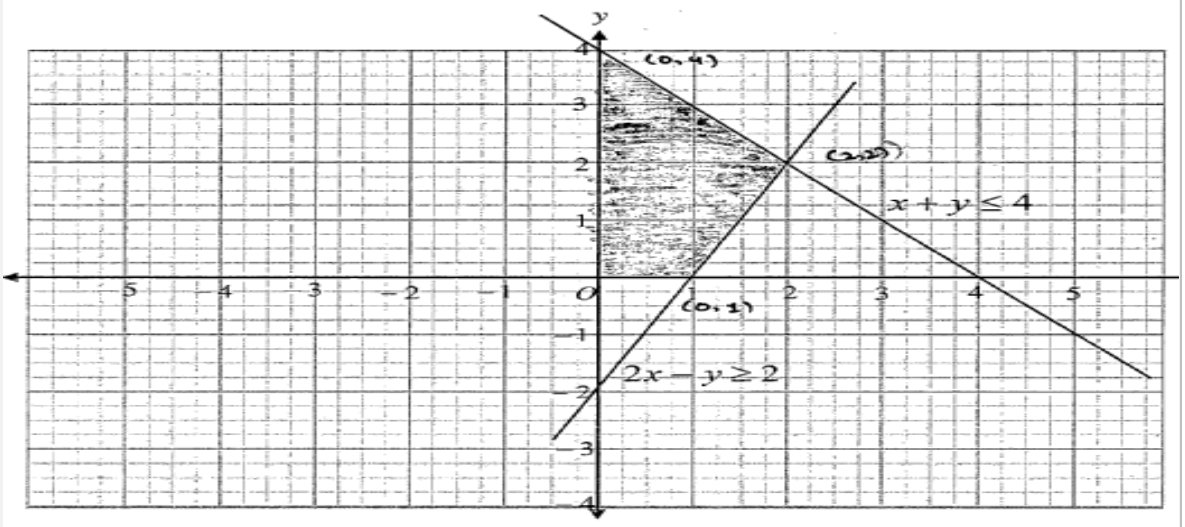
Description of Better Responses	In *better responses*, candidates correctly tested the given linear inequalities to determine whether they were true or false. They then accurately shaded the regions representing each inequality and successfully identified the common (feasible) region. Most students were able to find the intersecting area correctly and shade the region that satisfied both inequalities. After locating the feasible region, they correctly identified the corner points and presented a clear graphical representation.
Image of Better Response	 ii. identify all the corner points of the feasible region. (2 Marks) (1, 0), (2, 2), (4, 0)
Description of Weaker Responses	In *weaker responses*, candidates struggled to identify the feasible region because they misunderstood the conditions required by the inequalities. Some candidates shaded the individual inequalities incorrectly or failed to find the common region where both conditions were true. As a result, their graphs did not accurately show the feasible region. In several cases, candidates also identified incorrect corner points due to a poor understanding of where the boundary lines intersect. These errors led to an inaccurate graphical solution and incorrect final answers.
Images of Weaker Responses	**Image (i)**  ii. identify all the corner points of the feasible region. (2 Marks) $x+y\leq 4 = (0,4)(1,3),(2,2)(3,1)(4,0)(5,-1)$ $2x-y\geq 2 = (0,-2)(1,0)(2,2)$

Image (ii)



ii. identify all the corner points of the feasible region. (2)

(2, 2) , (0, 1) (0, 4)

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

Following are some practical, achievable, and classroom-friendly strategies and teaching methods to help improve students' understanding and performance in solving linear inequalities graphically, identifying the feasible region, and determining corner points:

Desmos or GeoGebra Graphing Practice: Use digital graphing tools so students can visualise and manipulate inequalities.

Activity: Give students inequalities and ask them to graph and adjust the shading until they find the feasible region.

Test a Point Technique: Teach students to use a test point (e.g., (0,0)) to check which side of the line to shade.

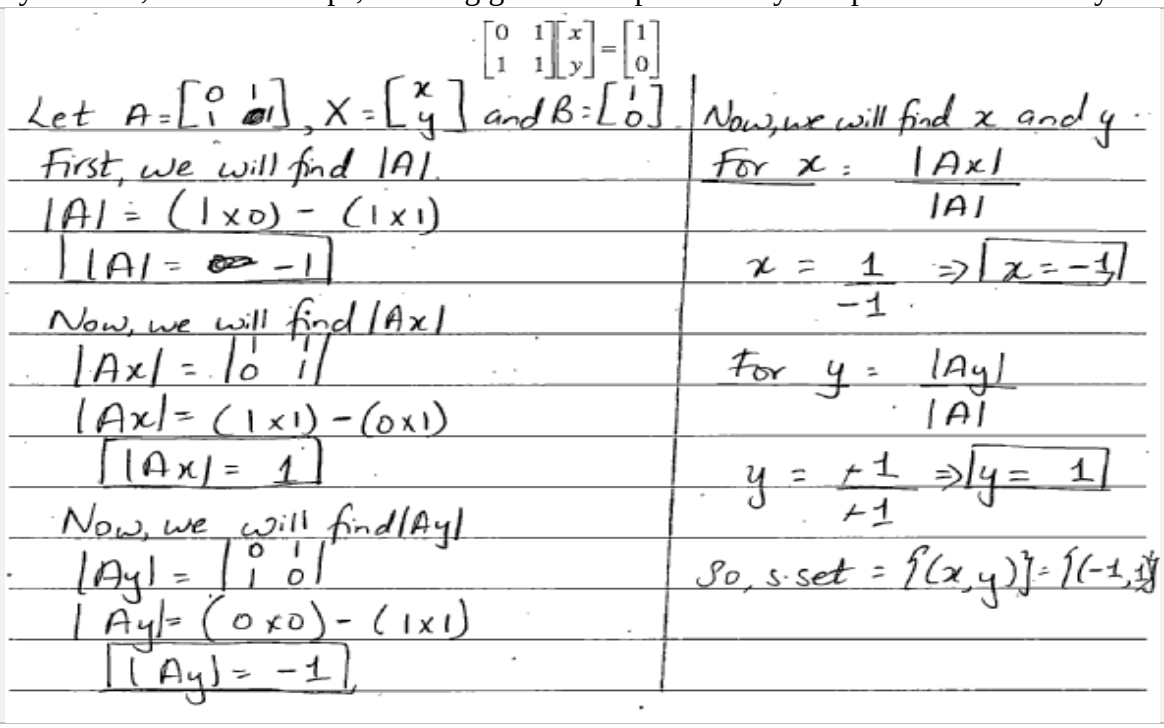
Activity: Give practice problems where students must test and explain which side satisfies each inequality.

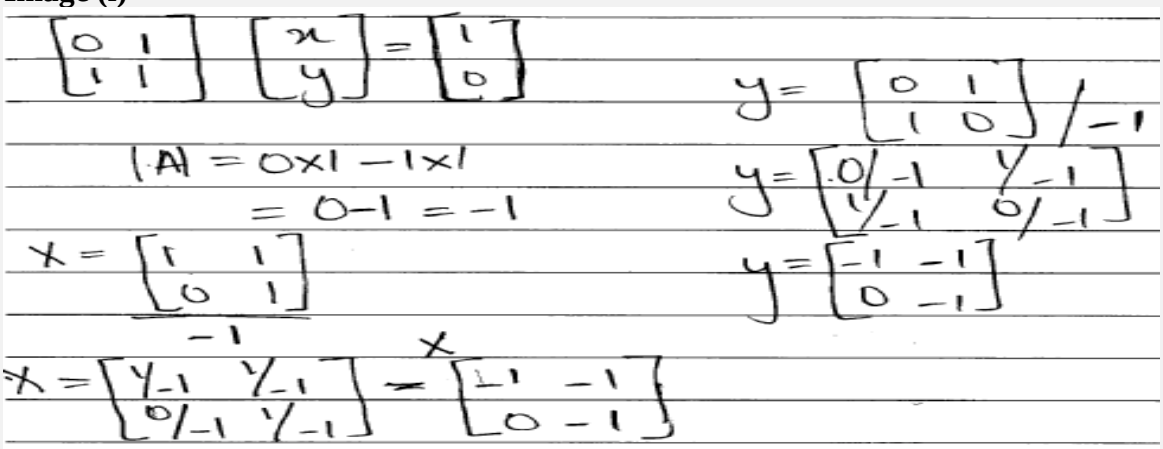
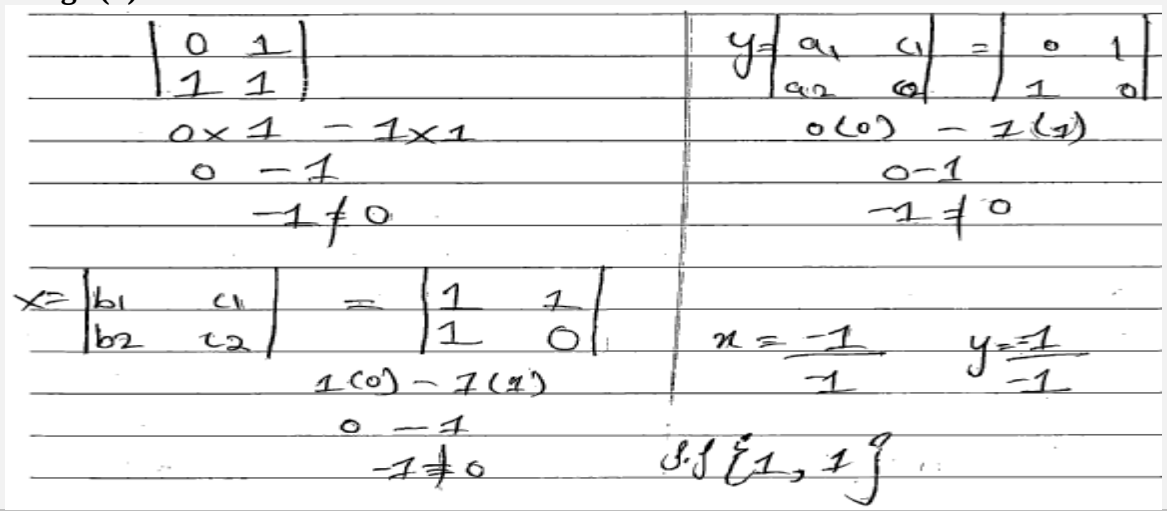
Feasible Region Puzzle: Give students cut-out graphs of individual inequalities.

Activity: Have them physically overlay or trace the graphs to find the common shaded region (the feasible area).

Spot the Error Task: Show incorrect graphs (wrong shading or wrong corner points).

Activity: Ask students to explain what's wrong and how they would fix it.

Question No. 5	
Question Text	Find the values of x and y by solving the given matrix equation using Cramer's rule. $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
SLO No.	6.4.1
SLO Text	Solve a matrix equation using the following methods: a. Inverse Matrix Method b. Cramer's rule
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for finding the determinant (main) 1 mark for getting $x = \frac{\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}}{-1} = (1 - 0)/ -1 = -1$ 1 mark for getting $y = \frac{\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}}{-1} = (0 - 1)/ -1 = 1$
Overall Performance	Overall, performance on this question was good. Many candidates applied Cramer's Rule correctly and followed a structured approach to solving simultaneous equations using matrices, though few students faced difficulties in forming the required matrices and calculating determinants.
Description of Better Responses	In *better responses*, candidates demonstrated a sound understanding of Cramer's Rule. They correctly identified and formed matrix A, and constructed matrix A_x and A_y using the given constants from the system of equations. The application of the rule was accurate, and candidates used determinants properly to find the values of x and y. Their working was systematic, with clear steps, showing good conceptual clarity and procedural accuracy.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates struggled with this question. A number of them were unable to construct the correct matrices A_x and A_y, which indicated a lack of understanding of the method. Others failed to apply the determinant formula correctly or made calculation errors while finding the values of x and y. These mistakes reflected weak grasp of the concept and led to incorrect solutions.

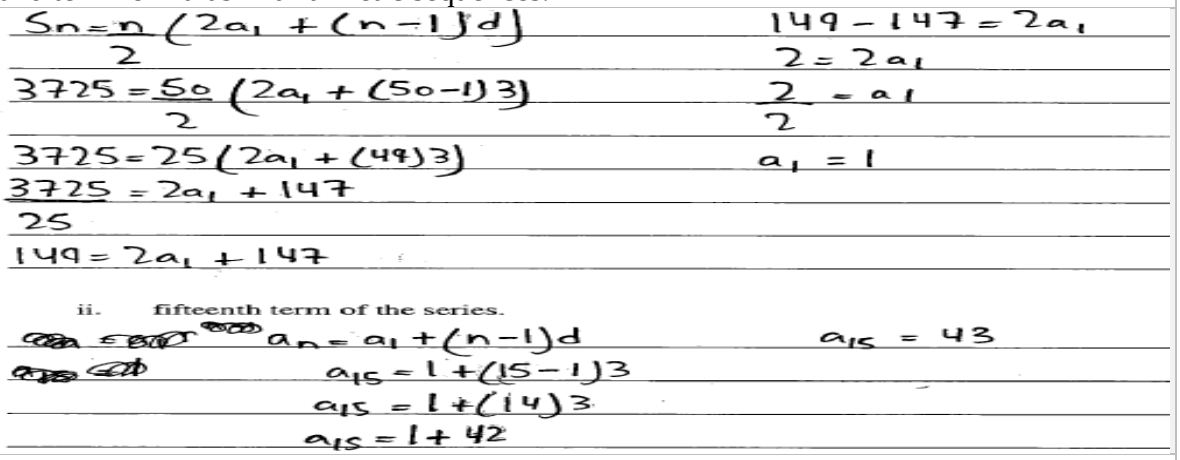
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

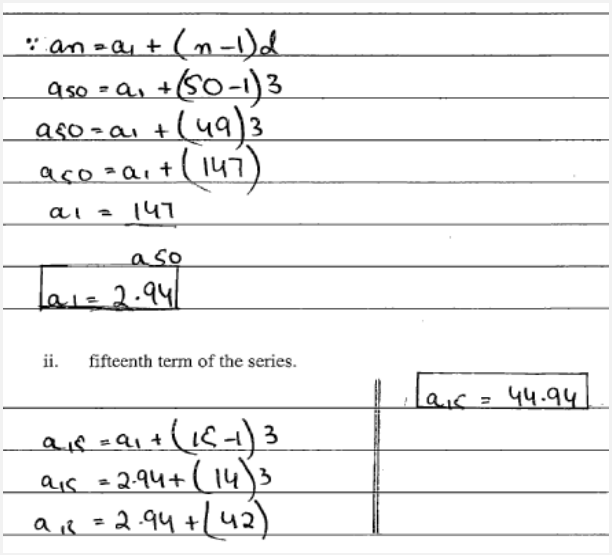
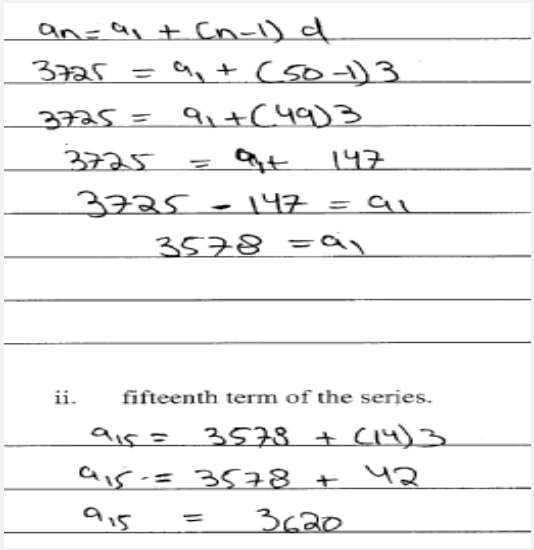
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion:
Following are some practical, achievable, and classroom-friendly strategies and teaching methods to help improve students' understanding and performance in solving Matrices and related problems:
Use of Equation Cards (Matching Game): Break down linear equations into separate components: coefficients of *x*, *y*, constants.
Activity: Give students shuffled cards for each component and ask them to correctly form matrix *A*, A_x, and A_y from them.

Hands-On Matrix Building with Placeholders: Use blank matrix grids with placeholders.
Activity: Write equations on the board, and students fill the blank grids with the appropriate values (coefficients or constants) to build matrices step by step.

Question No. 6a Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b.	
Question Text	The sum of 50 terms of an arithmetic sequence is 3,725. If the common difference of the series is 3, then find the i. first term of the series. ii. fifteenth term of the series.
SLO No.	8.2.2
SLO Text	Solve problems involving arithmetic series.
Max Marks	3
Cognitive Level	A
Checking Hints	i. 1 mark for substituting in the formula for arithmetic series 1 mark for calculation of the first term ii. 1 mark for finding the 15 term of the series using arithmetic progression formula
Overall Performance	This question was attempted as often as part (b). Many candidates applied the correct formulae and solved both parts step by step. However, some were confused about when and how to use arithmetic formulae. The main problems were choosing the right formula, setting up equations correctly, and simplifying algebraic expressions. This shows that while candidates may remember the formulae, they need more practice in applying and understanding them.
Description of Better Responses	In *better responses*, candidates demonstrated a clear understanding of arithmetic sequences by correctly applying the sum formula of an arithmetic series. They accurately identified the need to use the given number of terms, common difference, and total sum to set up a correct algebraic equation. After forming the equation, they simplified it appropriately and solved it to find the first term. In the second part, they used this value to apply the nth term formula correctly and calculated the 15th term of the sequence. The steps were shown clearly, with proper algebraic working, and the solutions were logically structured. These responses reflected strong procedural fluency and conceptual understanding of both the sum and term formulas in arithmetic sequences.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates struggled to understand the requirements of the question. Instead of using the sum formula, a common mistake was applying the nth term formula in the first part, which led to incorrect equations and confusion. Some candidates attempted to plug in values directly without forming or simplifying an equation, while others substituted incorrect values or used formulas in the wrong order. This reflected a lack of clarity in distinguishing between the use of the sum formula and the nth term formula in different parts of an arithmetic series problem.

Images of Weaker Responses	Image (i)	Image (ii)
		

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:
The following are some practical, achievable, and classroom-friendly strategies and teaching methods to help improve students' understanding and performance in solving arithmetic sequence problems:
Compare and Contrast Tasks: Give students two similar problems, one that requires the nth term formula and one that requires the sum formula—and ask them to identify the difference.
Common Mistake Spotting: Present a solution with errors (e.g., using the wrong formula, misplacing values), and have students identify and correct the mistake.
Formula Use Flowchart: Display a decision-making flowchart:
Do you need a specific term? → Use nth term formula
Do you need a total or sum? → Use sum formula

Question No. 6b

Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b.

Question Text	Aslam invested Rs 300,000 in a business. The capital grew at a growth rate of 9.04% in 10 years. Calculate the total worth of the capital after 10 years.
SLO No.	8.4.2
SLO Text	Find the sum to *n* terms of a finite geometric series.
Max Marks	3

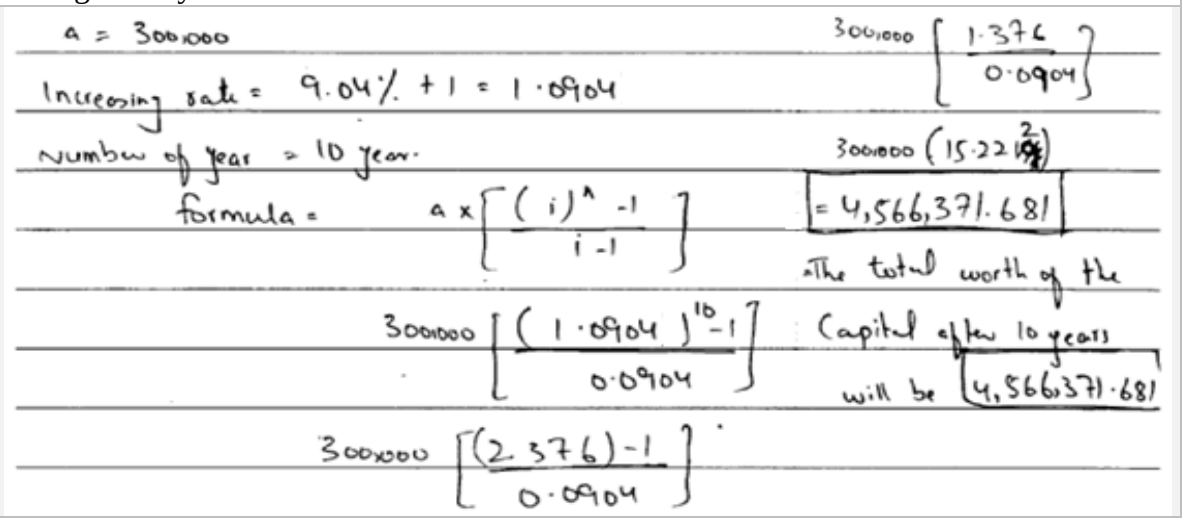
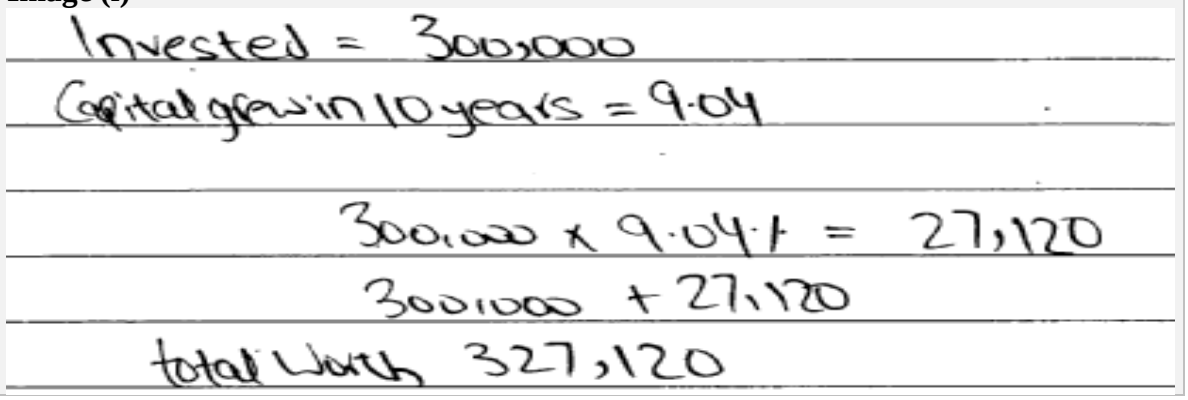
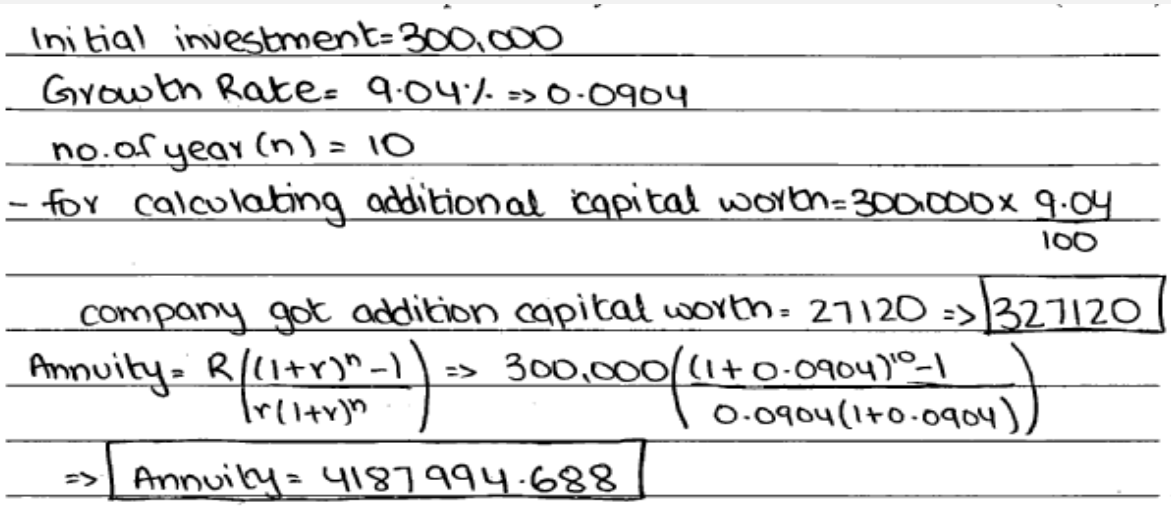
Cognitive Level	A
Checking Hints	1 mark for finding the value of r = 1.0904 1 mark for substitution in the correct formula ($S_n = \frac{a_1(r^n - 1)}{r - 1}$) 1 mark for finding the total amount
Overall Performance	Overall, performance on this question varied. While many candidates struggled to apply the geometric formula correctly and identify the key components, others showed confusion between arithmetic and geometric sequences. The most common errors involved incorrect formula selection, substitution mistakes, and misunderstanding the role of the common ratio in geometric progression.
Description of Better Responses	In *better responses*, candidates demonstrated strong comprehension by correctly identifying that the question involved a geometric series. They accurately recognised the first term (a), the common ratio (r), and the number of terms (n)**.** Using this information, they correctly applied the formula for the sum of a geometric series or the nth term as required. Their ability to select and substitute into the correct formula showed clear conceptual understanding and an ability to differentiate between arithmetic and geometric formulas based on the context of the question. The steps were well-structured, and their calculations were generally accurate.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates struggled to identify the correct type of series and often confused geometric series with arithmetic series. Many used the wrong formula, leading to incorrect substitution and results. Some candidates found the correct values of *a*, *r*, and *n* but either substituted them inaccurately or misapplied the formula. A few attempted to solve the question using the arithmetic sum formula instead of the geometric one, indicating a lack of clarity in distinguishing between the two types of sequences.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:
By implementing these strategies, educators can enhance students' understanding and application of complex concepts, ultimately improving overall performance on assessments.
Guided Identification Practice: Give students mixed examples and ask them only to identify:
Type of sequence (arithmetic or geometric), First term (a), common difference or ratio (d or r), and number of terms (n)
Real-Life Context Tasks: Use practical examples of geometric growth or decay (e.g., population growth, compound interest, or repeated halving) to introduce and apply the formula.
Think-Pair-Share on Formula Use: Pose a geometric problem, ask students to decide which formula to use, and explain their choice to a partner before solving.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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